

Q1. Title

Constraint-Based Problem Solving Using Backtracking Search

Problem Statement

A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) while satisfying the following constraints:

Constraints

1. D1 cannot work Night shift.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor can be assigned to each shift.
4. D3 cannot work Morning shift.
5. Every doctor must be assigned exactly one shift.

Objective

Apply the Backtracking Search Algorithm to assign doctors to shifts while satisfying all constraints. Show all intermediate constraint checks and determine the final valid schedule.

Q2. Title

Robot Path Planning using Breadth-First Search (BFS)

Problem Statement

A robot operates in a 5×5 grid environment. The robot starts from S (Start) and must reach G (Goal) while avoiding obstacles.

Constraints

1. Movement allowed: Up, Down, Left, Right
2. Cost of each move = 1
3. Robot cannot move through obstacles.
4. Use Manhattan Distance Heuristic.
5. Apply Breadth-First Search (BFS) to determine the optimal path.
6. Show step-by-step node expansion and queue updates.

Grid

S 0 X 0 0

0 0 X 0 0

0 0 0 0 X

X X 0 X 0

0 0 0 0 G

S = Start

G = Goal

X = Obstacle

0 = Free Cell

Q3. Title

Autonomous Rescue Robot using Uniform Cost Search (UCS)

Problem Statement

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot starts at S (Start) and must reach G (Goal) while avoiding blocked paths and minimizing the total cost.

Constraints

1. The robot can move Up, Down, Left, Right.
2. Some cells are blocked (obstacles).
3. The robot can only observe adjacent cells (limited sensing).
4. Each movement has a cost of 1.
5. Entering a Risk Zone (R) adds an additional cost of 2.
6. The robot must minimize the total cost.
7. The robot updates its knowledge dynamically (Online Search).

Grid

S 0 R X 0

0 X 0 0 0

0 0 X R 0

R 0 0 X 0

0 0 0 0 G

Legend

- S = Start
- G = Goal
- X = Obstacle
- R = Risk Zone
- 0 = Safe Cell